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How to Read Science and Mathematics

The title of this chapter may be misleading. We do not propose to give you advice about how to read every kind of science and mathematics. We will confine ourselves to discussing only two kinds: the great scientific and mathematical classics of our tradition, on the one hand, and modern scientific popularizations, on the other hand. What we say will often be applicable to the reading of specialized monographs on abstruse and limited subjects, but we cannot help you to read those. There are two reasons for this. One is, simply, that we are not qualified to do it.

The other is this. Until approximately the end of the nineteenth century, the major scientific books were written for a lay audience. Their authors—men like Galileo, and Newton, and Darwin—were not averse to being read by specialists in their fields; indeed, they wanted to reach such readers. But there was as yet no institutionalized specialization in those days, days which Albert Einstein called “the happy childhood of science.” Intelligent and well-read persons were expected to read scientific books as well as history and philosophy; there were no hard and fast distinctions, no boundaries that could not be crossed. There was also none of the disregard for the general or lay reader that is manifest in contemporary scientific writing. Most modern scientists do not care what lay readers think, and so they do not even try to reach them.

Today, science tends to be written by experts for experts. A serious communication on a scientific subject assumes so much specialized knowledge on the part of the reader that it usually cannot be read at all by anyone not learned in the field. There are obvious advantages to this approach, not least that it serves to advance science more quickly. Experts talking to each other about their expertise can arrive very quickly at the frontiers of it—they can see the problems at once and begin to try to solve them. But the cost is equally obvious. You—the ordinary intelligent reader whom we are addressing in this book—are left quite out of the picture.

In fact, this situation, although it is more extreme in science than elsewhere, obtains in many other fields as well. Nowadays, philosophers seldom write for anyone except other philosophers; economists write for economists; and even historians are beginning to find that the kind of shorthand, monographic communication to other experts that has long been dominant in science is a more convenient way of getting ideas across than the more traditional narrative work written for everyone.

What does the general reader do in these circumstances? He cannot become expert in all fields. He must fall back, therefore, on scientific popularizations. Some of these are good, and some are bad. But it is not only important to know the difference; it is also important to be able to read the good ones with understanding.

Understanding the Scientific Enterprise

One of the fastest growing academic disciplines is the history of science. We have seen marked changes in this area within the past few years. It was not so long ago that “serious” scientists looked down upon historians of science.

The latter were thought of as men who studied the history of a subject because they were not capable of expanding its frontiers. The attitude of scientists to historians of science could be summed up in that famous remark of George Bernard Shaw's: "Those who can, do; those who can't, teach."

Expressions of this attitude are seldom heard nowadays. Departments of the history of science have become respectable, and excellent scientists study and write about the history of their subject. An example is what has been called the "Newton industry." At the present time, intensive and extensive research is being undertaken in many countries on the work and strange personality of Sir Isaac Newton. Half a dozen books have been recently published or announced. The reason is that scientists are more concerned than ever before about the nature of the scientific enterprise itself.

Thus we have no hesitation in recommending that you try to read at least some of the great scientific classics of our tradition. In fact, there is really no excuse for not **trying** to read them. None of them is impossibly difficult, not even a book like Newton's ***Mathematical Principles of Natural Philosophy***, if you are willing to make the effort.

The most helpful advice we can give you is this. You are required by one of the rules for reading expository works to state, as clearly as you can, the problem that the author has tried to solve. This rule of analytical reading is relevant to all expository works, but ***it is particularly relevant to works in the fields of science and mathematics.***

There is another way of saying this. As a layman, you do not read the classical scientific books to become knowledgeable in their subject matters in a contemporary sense. Instead, you read them to understand the history and philosophy of science. That, indeed, is the layman's responsibility with regard to science. The major way in which you can discharge it is to become aware of the problems that the great scientists were trying to solve—aware of the problems, and aware, also, of the background of the problems.

To follow the strands of scientific development, to trace the ways in which facts, assumptions, principles, and proofs are interrelated, is to engage in the activity of the human reason where it has probably operated with the most success. That is enough by itself, perhaps, to justify the historical study of science. In addition, such study will serve to dispel, in some measure, the apparent unintelligibility of science. Most important of all, it is an activity of the mind that is essential to education, the central aim of which has always been recognized, from Socrates' day down to our own, as the freeing of the mind through the discipline of wonder.

Suggestions for Reading Classical Scientific Books

By a scientific book, we mean the report of findings or conclusions in some field of research, whether carried on experimentally in a laboratory or by observations of nature in the raw. The scientific problem is always to describe the phenomena as accurately as possible, and to trace the interconnections between different kinds of phenomena.

In the great works of science, there is no oratory or propaganda, though there may be bias in the sense of initial presuppositions. You detect this, and take account of it, by distinguishing what the author **assumes** from what he **establishes** through argument. The more "objective" a scientific author is, the more he will explicitly ask you to take this or that for granted. Scientific objectivity is not the **absence of initial bias**. It is attained by **frank confession of it**.

The leading terms in a scientific work are usually expressed by uncommon or technical words. They are relatively easy to spot, and through them you can readily grasp the propositions. The main propositions are always general ones. Science thus is not chronotopic. Just the opposite; a scientist, unlike a historian, tries to get away from locality in time and place. He tries to say how things are generally, how things generally behave.

There are likely to be two main difficulties in reading a scientific book. One is with respect to the arguments. Science is primarily inductive; that is, its primary arguments are those that establish a general proposition by reference to observable evidence—a single case created by an experiment, or a vast array of cases collected by patient investigation. There are other arguments, of the sort that are called deductive. These are arguments in which a proposition is proved by other propositions already somehow established. So far as proof is concerned, science does not differ much from philosophy. But the inductive argument is characteristic of science.

This first difficulty arises because, in order to understand the inductive arguments in a scientific book, you must be able to follow the evidence that the scientist reports as their basis. Unfortunately, this is not always possible with nothing but the book in hand. If the book itself fails to enlighten him, the reader has only one recourse, which is to get the necessary special experience for himself at first hand. He may have to witness a laboratory demonstration. He may have to look at and handle pieces of apparatus similar to those referred to in the book. He may have to go to a museum and observe specimens or models.

Anyone who desires to acquire an understanding of the history of science must not only read the classical texts, but must also become acquainted, through direct experience, with the crucial experiments in that history. There are classical experiments as well as classical books. The scientific classics become more intelligible to those who have seen with their own eyes and done with their own hands what a great scientist describes as the procedure by which he reached his insights.

This does not mean that you cannot make a start without going through all the steps described. Take a book like Lavoisier's *Elements of Chemistry*, for instance. Published in 1789, the work is no longer considered to be useful as a textbook in chemistry, and indeed a student would be unwise to study it for the purpose of passing even a high school examination in the subject. Nevertheless, its method was revolutionary at the time, and its conception of a chemical element is still, on the whole, the one that we have in modern times. Now the point is that you do not have to read the book through, and in detail, to receive these impressions of it. The Preface, for example, with its emphasis on the importance of method in science, is enlightening. "Every branch of physical science," wrote Lavoisier,

must consist of three things: the series of facts which are the objects of the science, the ideas which represent these facts, and the words by which these facts are expressed. . . . And, as ideas are preserved and communicated by means of words, it necessarily follows that we cannot improve the language of any science without at the same time improving the science itself; neither can we, on the other hand, improve a science without improving the language or nomenclature which belongs to it.

This was exactly what Lavoisier did. He improved chemistry by improving its language, just as Newton, a century before, had improved physics by systematizing and ordering *its* language—in the process, as you may recall, developing the differential and integral calculus.

Mention of the calculus leads us to consider the second main difficulty in reading scientific books. And that is the problem of mathematics.

Facing the Problem of Mathematics

Many people are frightened of mathematics and think they cannot read it at all. No one is quite sure why this is so. Some psychologists think there is such a thing as "symbol blindness"—the inability to set aside one's

dependence on the concrete and to follow the controlled shifting of symbols. There may be something to this, except, of course, that words shift, too, and their shifts, being more or less uncontrolled, are perhaps even more difficult to follow. Others believe that the trouble lies in the teaching of mathematics. If so, we can be gratified that much recent research has been devoted to the question of how to teach it better.

The problem is partly this. We are not told, or not told early enough so that it sinks in, that mathematics is a language, and that we can learn it like any other, including our own. We have to learn our own language twice, first when we learn to speak it, second when we learn to read it. Fortunately, mathematics has to be learned only once, since it is almost wholly a written language.

As we have already observed, learning a new written language always involves us in problems of elementary reading. When we underwent our initial reading instruction in elementary school, our problem was to learn to recognize certain arbitrary symbols when they appeared on a page, and to memorize certain relations among these symbols. Even the best readers continue to read, at least occasionally, at the elementary level: for example, whenever we come upon a word that we do not know and have to look up in the dictionary. If we are puzzled by the syntax of a sentence, we are also working at the elementary level. Only when we have solved these problems can we go on to read at higher levels.

Since mathematics is a language, it has its own vocabulary, grammar, and syntax, and these have to be learned by the beginning reader. Certain symbols and relationships between symbols have to be memorized. The problem is different, because the language is different, but it is no more difficult, **theoretically**, than learning to read English or French or German. At the elementary level, in fact, it may even be easier.

Any language is a medium of communication among men on subjects that the communicants can mutually comprehend. The subjects of ordinary discourse are mainly emotional facts and relations. Such subjects are not **entirely** comprehensible by any two different persons. But two different persons **can** comprehend a third thing that is outside of and emotionally separated from both of them, such as an electrical circuit, an isosceles triangle, or a syllogism. It is mainly when we invest these things with emotional connotations that we have trouble understanding them. Mathematics allows us to avoid this. There are no emotional connotations of mathematical terms, propositions, and equations when these are properly used.

We are also not told, at least not early enough, how beautiful and how intellectually satisfying mathematics can be. It is probably not too late for anyone to see this if he will go to a little trouble. You might start with Euclid, whose ***Elements of Geometry*** is one of the most lucid and beautiful works of any kind that has ever been written.

Let us consider, for example, the first five propositions in Book I of the ***Elements***. (If a copy of the book is available, you should look at it.) Propositions in elementary geometry are of two kinds: (1) the statement of problems in the construction of figures, and (2) theorems about the relations between figures or their parts. Construction problems require that something be done, theorems require that something be proved. At the end of a Euclidean construction problem, you will find the letters Q.E.F., which stand for ***Quod erat faciendum***, “(Being) what it was required to do.” At the end of a Euclidean theorem, you will find the letters Q.E.D., which stand for ***Quod erat demonstrandum***, “(Being) what it was required to prove.”

The first three propositions in Book I of the ***Elements*** are all problems of construction. Why is this? One answer is that the constructions are needed in the proofs of the theorems. This is not apparent in the first four propositions, but we can see it in the fifth proposition, which is a theorem. It states that in an isosceles triangle (a triangle with two equal sides) the base angles are equal. This involves the use of Proposition 3, for a shorter line is cut off from a longer line. Since Proposition 3, in turn, depends on the use of the construction in Proposition 2, while Proposition 2 involves Proposition 1, we see that these three constructions are needed for the sake of Proposition 5.

Constructions can also be interpreted as serving another purpose. They bear an obvious similarity to postulates; both constructions and postulates assert that geometrical operations can be performed. In the case of the postulates, the possibility is **assumed**; in the case of the propositions, it is **proved**. The proof, of course, involves the use of the postulates. Thus, we might wonder, for example, whether there is really any such thing as an equilateral triangle, which is defined in Definition 20. Without troubling ourselves here about the thorny question of the existence of mathematical objects, we can at least see that Proposition 1 shows that, from the **assumption** that there are such things as straight lines and circles, it **follows** that there are such things as equilateral triangles.

Let us return to Proposition 5, the theorem about the equality of the base angles of an isosceles triangle. When the conclusion has been reached, in a series of steps involving reference to previous propositions and to the postulates, the proposition has been proved. It has then been shown that **if something is true** (namely, the hypothesis that we have an isosceles triangle), and if some additional things are valid (the definitions, postulates, and prior propositions), **then something else is also true**, namely, the conclusion. The proposition asserts this **if-then** relationship. It does not assert the truth of the hypothesis, nor does it assert the truth of the conclusion, except when the hypothesis is true. Nor is this connection between hypothesis and conclusion seen to be true until the proposition is proved. It is precisely the truth of this connection that is proved, and nothing else.

Is it an exaggeration to say that this is beautiful? We do not think so. What we have here is a **really logical exposition** of a **really limited problem**. There is something very attractive about both the clarity of the exposition and the limited nature of the problem. Ordinary discourse, even very good philosophical discourse, finds it difficult to limit its problems in this way. And the use of logic in the case of philosophical problems is hardly ever as clear as this.

Consider the difference between the argument of Proposition 5, as outlined here, and even the simplest of syllogisms, such as the following:

All animals are mortal;
All men are animals;
Therefore, all men are mortal.

There is something satisfying about that, too. We can treat it as though it were a piece of mathematical reasoning. Assuming that there are such things as animals and men, and that animals are mortal, then the conclusion follows with the same certainty as the one about the angles of the triangle. But the trouble is that there really are animals and men; we are assuming something about real things, something that may or may not be true. We have to examine our assumptions in a way that we do not have to do in mathematics. Euclid's proposition does not suffer from this. It does not really matter to him whether there are such things as isosceles triangles. **If** there are, he is saying, and **if** they are defined in such and such a way, **then** it follows absolutely that their base angles are equal. There can be no doubt about this whatever—now and forever.

Handling the Mathematics in Scientific Books

This digression on Euclid has led us a little out of our way. We were observing that the presence of mathematics in scientific books is one of the main obstacles to reading them. There are a couple of things to say about that.

First, you can probably read at least elementary mathematics better than you think. We have already suggested that you should begin with Euclid, and we are confident that if you spent several evenings with the **Elements** you

would overcome much of your fear of the subject. Having done some work on Euclid, you might proceed to glance at the works of other classical Greek mathematicians—Archimedes, Apollonius, Nicomachus. They are not really very difficult, and besides, you can skip.

That leads to the second point we want to make. If your intention is to read **a mathematical book in and for itself**, you must read it, of course, from beginning to end—and with a pencil in your hand, for writing in the margins and even on a scratch pad is more necessary here than in the case of any other kinds of books. But your intention may not be that, but instead to read **a scientific work that has mathematics in it**. In this case, skipping is often the better part of valor.

Take Newton's **Principia** for an example. The book contains many propositions, both construction problems and theorems, but it is not necessary to read all of them in detail, especially the first time through. Read the statement of the proposition, and glance down the proof to get an idea of how it is done; read the statements of the so-called lemmas and corollaries; and read the so-called scholiums, which are essentially discussions of the relations between propositions and of their relations to the work as a whole. You will begin to see that whole if you do this, and so to discover how the system that Newton is constructing is built—what comes first and what second, and how the parts fit together. Go through the whole work in this way, avoiding the diagrams if they trouble you (as they do many readers), merely glancing at much of the interstitial matter, but being sure to find and read the passages where Newton is making his main points. One of these comes at the very end of the work, at the close of Book III, which is titled "The System of the World." This **General Scholium**, as Newton called it, not only sums up what has gone before but also states the great problem of almost all subsequent physics.

Newton's **Optics** is another scientific classic that you might want to try to read. There is actually very little mathematics in it, although at first glance that does not appear to be so because the pages are sprinkled with diagrams. But these diagrams are merely illustrations describing Newton's experiments with holes for the sun to shine through into a dark room, with prisms to intercept the sunbeam, and with pieces of white paper placed so that the various colors of the beam can shine on them. You can quite easily repeat some of these experiments yourself, and this is fun to do, for the colors are beautiful, and the descriptions are eminently clear. You will want to read, in addition to the descriptions of the experiments, the statements of the various theorems or propositions, and the discussions that occur at the end of each of the three Books, where Newton sums up his discoveries and suggests their consequences. The end of Book III is famous, for it contains some statements by Newton about the scientific enterprise itself that are well worth reading.

Mathematics is very often employed by scientific writers, mainly because it has the qualities of preciseness, clarity, and limitedness that we have described. Usually you can understand something of the matter without going very deeply into the mathematics, as in the case of Newton. Oddly enough, however, even if mathematics is absolutely terrifying to you, its absence from certain works may cause you even more trouble. A case in point is Galileo's **Two New Sciences**, his famous treatise on the strength of materials and on motion. This work is particularly difficult for modern readers because it is **not** primarily mathematical; instead, it is presented in the form of a dialogue. The dialogue form, though appropriate to the stage and useful in philosophy when employed by such a master as Plato, is not really appropriate to science. It is therefore hard to discover what Galileo is saying, although when you do you will discover that he is stating some revolutionary things.

Not all of the scientific classics, of course, employ mathematics or even need to employ it. The works of Hippocrates, the founder of Greek medicine, are not mathematical. You might well read them to discover Hippocrates' view of medicine—namely, that it is the art of keeping people well, rather than that of curing them when they are sick. That is unfortunately an uncommon idea nowadays. Nor is William Harvey's discourse on the circulation of the blood mathematical, or William Gilbert's book on magnets. They can be read without too

much difficulty if you always keep in mind that your primary obligation is **not to become competent in the subject matter but instead to understand the problem.**

A Note on Popular Science

In a sense, there is little more to say about reading scientific popularizations. By definition, these are works—either books or articles—written for a wide audience, not just for specialists. Thus, if you have managed to read some of the classics of the scientific tradition, you should not have much trouble with them. This is because, although they are about science, they generally skirt or avoid the two main problems that confront the reader of an original contribution in science. First, **they contain relatively few descriptions of experiments** (instead, they merely report the results of experiments). Second, **they contain relatively little mathematics** (unless they are popular books about mathematics itself).

Popular scientific articles are usually easier to read than popular scientific books, although not always. Sometimes such articles are very good—for example, articles found in **Scientific American**, a monthly magazine, or **Science**, a somewhat more technical weekly publication. Of course, these publications, no matter how good they are or how carefully and responsibly edited, pose the problem that was discussed at the end of the last chapter. In reading them, we are at the mercy of reporters who filter the information for us. If they are good reporters, we are fortunate. If they are not, we have almost no recourse.

Scientific popularizations are never easy reading in the sense that a story is or seems to be. Even a three-page article on DNA containing no reports of experiments and no diagrams or mathematical formulas demands considerable effort on the part of the reader. You cannot read it for understanding without keeping your mind awake. Thus, the requirement that you read actively is more important here than almost anywhere else. Identify the subject matter. Discover the relation between the whole and its parts. Come to terms and plot the propositions and arguments. Work at achieving understanding before you begin to criticize or to assess significance. These rules, by now, are all familiar. But they apply here with particular force.

Short articles are usually primarily informational, and as such they require less active thinking on your part. You must make an effort to understand, to follow the account provided by the author, but you often do not have to go beyond that. In the case of such excellent popular books as Whitehead's **Introduction to Mathematics**, Lincoln Barnett's **The Universe and Dr. Einstein**, and Barry Commoner's **The Closing Circle**, something more is required. This is particularly true of a book like Commoner's, on a subject—the environmental crisis—of special interest and importance to all of us today. The writing is compact and requires constant attention. But the book as a whole has implications that the careful reader will not miss. Although it is not a practical work, in the sense described above in Chapter 13, its theoretical conclusions have important consequences. The mere mention of the book's subject matter—the environmental crisis—suggests this. The environment in question is our own; if it is undergoing a crisis of some sort, then it inevitably follows, even if the author had not said so—though in fact he has—that we are also involved in the crisis. The thing to do in a crisis is (usually) to act in a certain way, or to stop acting in a certain way. Thus Commoner's book, though essentially theoretical, has a significance that goes beyond the theoretical and into the realm of the practical.

This is not to suggest that Commoner's work is important and the books by Whitehead and Barnett unimportant. When **The Universe and Dr. Einstein** was written, as a theoretical account (written for a popular audience) of the history of researches into the atom, people were widely aware of the perils inherent in atomic physics, as represented mainly but not exclusively by the recently discovered atomic bomb. Thus that theoretical book also had practical consequences. But even if people are today not so worried about the imminence of an

atomic or nuclear war, there is still what may be called a practical necessity to read this theoretical book, or one like it. The reason is that atomic and nuclear physics is one of the great achievements of our age. It promises great things for man, at the same time that it poses great perils. An informed and concerned reader should know everything he can about the subject.

A slightly different urgency is exerted by Whitehead's ***Introduction to Mathematics***. Mathematics is one of the major modern mysteries. Perhaps it is the leading one, occupying a place in our society similar to the religious mysteries of another age. If we want to know something about what our age is all about, we should have some understanding of what mathematics is, and of how the mathematician operates and thinks. Whitehead's book, although it does not go very deeply into the more abstruse branches of the subject, is remarkably eloquent about the principles of mathematical reasoning. If it does nothing else, it shows the attentive reader that the mathematician is an ordinary man, not a magician. And that discovery, too, is important for any reader who desires to expand his horizons beyond the immediate here and now of thought and experience.